

Generative AI Chatbot for Student Assistance

Team Member 1 Harsh Shukla (92410103138)

Team Member 2 Shyam Patel (92410103117)

Team Member 3 Param Sanghvi (92410103130)

Prof. Nitin Varshney of Internal Guide

Abstract

The Indian higher education landscape is immensely complex. Millions of graduating students must make critical academic decisions based on fragmented data like tuition fees and placement records. While modern Large Language Models (LLMs) offer unprecedented conversational fluency, deploying them directly into educational counseling introduces a critical flaw: AI Hallucination. Traditional LLMs rely on parametric weights rather than real-time databases, frequently inventing fabricated numerical data when queried about specific local institutions. In financial planning, a hallucinated tuition fee is a deployment failure that can severely mislead vulnerable students. To resolve this crisis, this research proposes a localized Retrieval-Augmented Generation (RAG) system designed to absolutely refuse to hallucinate by decoupling knowledge storage from language generation. The methodology began by curating a dense, 18,000-cell relational matrix encompassing multidimensional profiles for 300 Indian universities. To transition this tabular data into searchable mathematical space, the architecture deployed the all-MiniLM-L6-v2 semantic embedding model, mapping the dataset into a 384-dimensional Facebook AI Similarity Search (FAISS) vector database. For the generative backbone, the system utilized the open-weights Qwen2.5-1.5B-Instruct LLM, quantized to 16-bit precision for efficient local deployment. To enforce a zero-hallucination rate, greedy decoding was activated alongside strict prompt guardrails. These guardrails bypassed the LLM's parametric memory, forcing it to act exclusively as a deterministic text-formatting engine for the FAISS-retrieved context. Empirical validation was conducted using an automated 1,500-question testing suite interspersed with adversarial trick queries. The quantitative results were exceptional: FAISS achieved 100.00% Retrieval Recall, and the LLM achieved 98.68% Precision, culminating in a 99.34% F1-Score. A Receiver Operating Characteristic (ROC) analysis yielded an Area Under the Curve (AUC) of 0.967, proving the system's ability to mathematically distinguish valid queries from conversational noise. Ultimately, this architecture establishes a new quantitative benchmark for deploying secure, hyper-accurate Generative AI in the educational sector.